AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_FDCC975D2B
Taxa	Oviraptor
Clade	Theropod
PBDB Records	3
Mass (regression)	138.6 kg (Campione & Evans 2012)
Mass (volumetric)	136.9 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.161 m^3
Validity Score	1.0000 / 1.0000
TCI	0.7657 (partial)
Axiom I Entropy	0.9768
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.8835
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.07, 0.201, 0.0]
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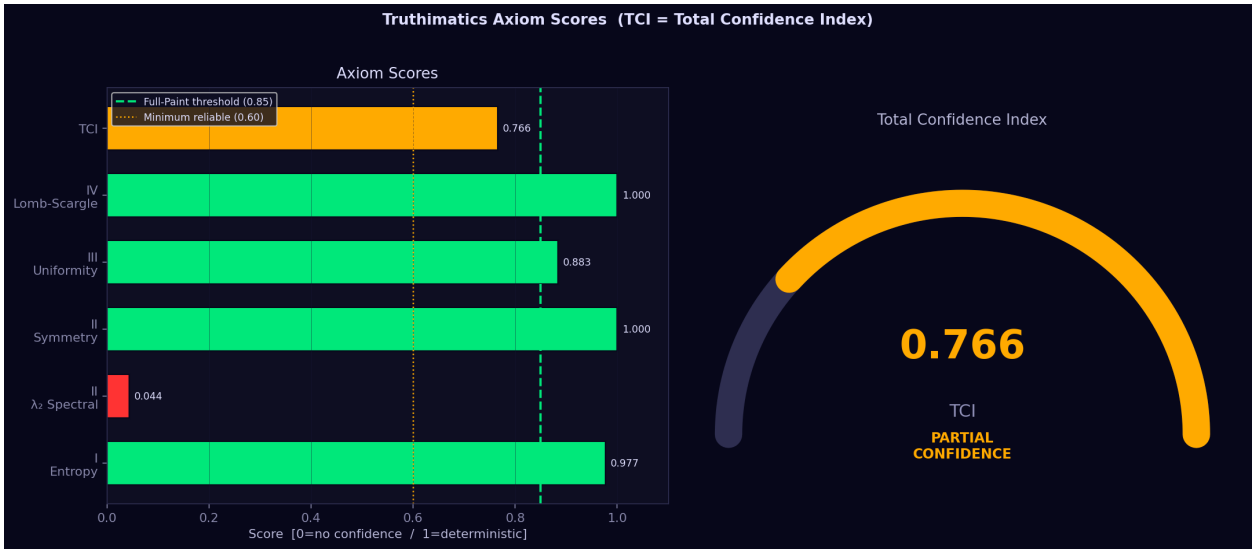
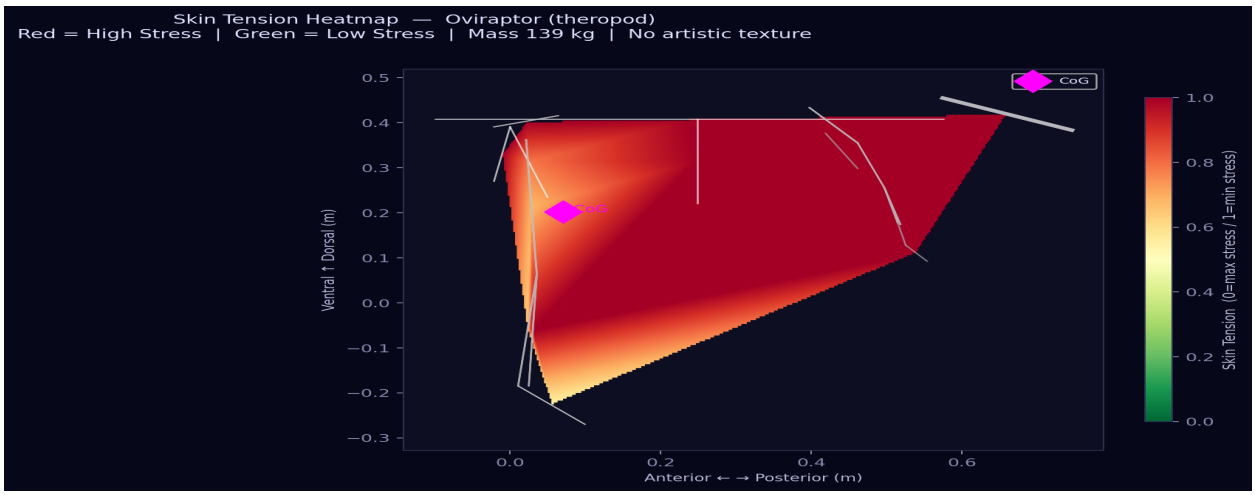
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



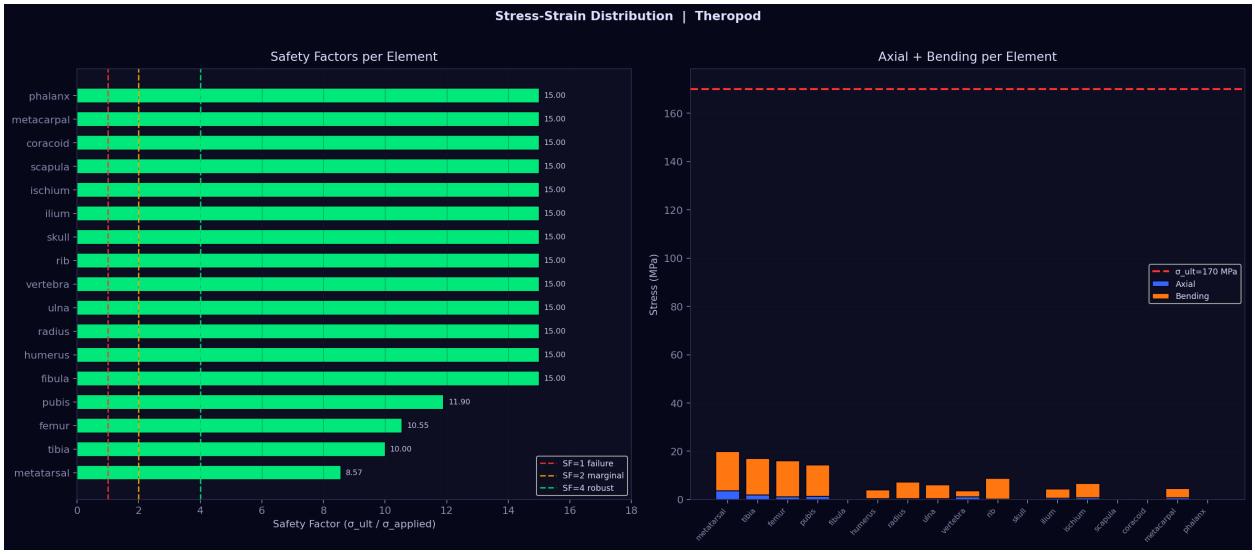
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

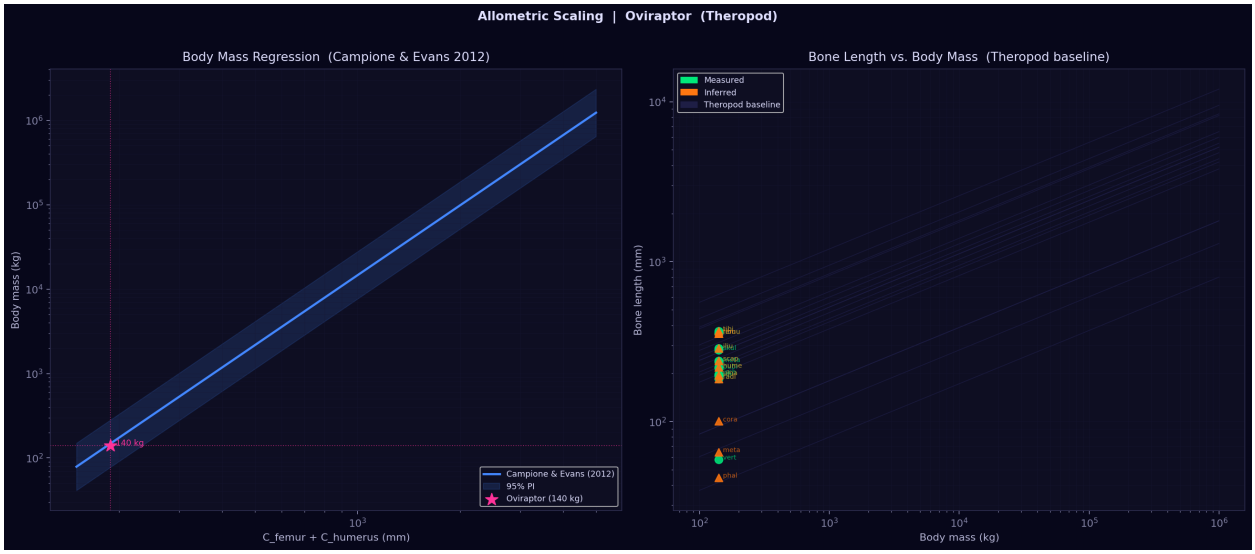


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.015	0.035	0.049	15.000	OK
femur	1.190	14.920	16.109	10.553	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.284	3.777	4.061	15.000	OK
ilium	0.635	3.779	4.414	15.000	OK
ischium	0.767	5.824	6.591	15.000	OK
metacarpal	0.727	3.873	4.600	15.000	OK
metatarsal	3.550	16.287	19.836	8.570	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.242	13.038	14.280	11.904	OK
radius	0.334	6.913	7.247	15.000	OK
rib	0.224	8.648	8.872	15.000	OK
scapula	0.009	0.042	0.051	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	1.909	15.087	16.996	10.002	OK
ulna	0.285	5.731	6.016	15.000	OK
vertebra	1.190	2.438	3.627	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.958 m	Femur + tibia + metatarsal
Estimated Speed	9.6 km/h (2.67 m/s)	Gait: Trot | Fr=0.761
Stride Length	2.024 m	Alexander (1976) / Thulborn (1990)
Bite Force	188.7 N (19.2 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	280.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3363	Circumference / length (dimensionless)
Limb Strength Index	3.4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	3.21 m	Allometric from femur length
Est. Neck Length	0.45 m	Allometric from femur length
Est. Tail Length	1.22 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	100.5	57.2	131.95	164.7	Constraint	0.40	15.00
femur	355.0	38.0	205.33	256.3	Measured	0.88	10.55
femur_right	355.0	38.0	205.33	256.3	Symmetry	0.84	10.55
fibula	355.0	10.0	10.04	10.5	Measured	0.88	15.00
fibula_right	355.0	10.0	10.04	10.5	Symmetry	0.84	15.00
humerus	218.0	22.0	42.26	52.8	Measured	0.88	15.00
humerus_right	218.0	22.0	42.26	52.8	Symmetry	0.84	15.00
ilium	288.0	52.0	311.93	389.4	Measured	0.88	15.00
ilium_right	288.0	52.0	311.93	389.4	Symmetry	0.84	15.00
ischium	198.0	28.0	62.18	77.6	Measured	0.88	15.00
metacarpal	64.3	13.3	1.71	1.4	Constraint	0.65	15.00
metatarsal	235.0	22.0	45.56	56.9	Measured	0.88	8.57
phalanx	44.7	11.9	0.77	0.6	Constraint	0.40	15.00
pubis	215.0	22.0	41.68	52.0	Measured	0.88	11.90
radius	185.0	12.0	10.67	13.3	Measured	0.88	15.00
radius_right	185.0	12.0	10.67	13.3	Symmetry	0.84	15.00
rib	192.0	8.0	4.92	6.1	Measured	0.88	15.00
scapula	240.0	72.0	498.35	622.1	Measured	0.88	15.00
scapula_right	240.0	72.0	498.35	622.1	Symmetry	0.84	15.00
skull	280.0	190.0	4048.79	5054.1	Measured	0.88	15.00
tibia	368.0	30.0	132.66	165.6	Measured	0.88	10.00
tibia_right	368.0	30.0	132.66	165.6	Symmetry	0.84	10.00
ulna	195.0	13.0	13.20	16.5	Measured	0.88	15.00
ulna_right	195.0	13.0	13.20	16.5	Symmetry	0.84	15.00
vertebra	58.0	38.0	33.55	41.9	Measured	0.88	15.00